GlycoTwin

An evolving patient digital twin for predictive diabetes management

Forecasts glucose at 30, 60 and 120 minutes, estimates the chance of crossing 180 mg/dL within the hour, and shows a clinician what drove the estimate.

> Research prototype on synthetic data. Not a medical device, not a diagnostic tool, not a substitute for clinical care. No medication or dosing guidance.

`glycotwin-0.1.0` `trained 2026-10-23`

The problem

Current monitoring looks backwards

A glucose reading tells a clinician where a patient is now. It does not say whether they are heading for an excursion, how fast, or why.

- Clinical records and wearable measurements sit in separate systems
- A single reading carries no temporal context: rising fast or levelling off
- Population thresholds miss what is normal for one individual
- No unified place to ask what a change in behaviour would do to the outlook

> Can historical clinical data and live physiological signals be combined into an evolving twin that predicts a crossing before it happens, and explains itself?

What we built

A twin that updates on every reading

1. **Validate** units, range, duplicates, gaps, late and out of order arrival
2. **Update** a rolling 4 hour window and the patient's twin state
3. **Forecast** glucose at three horizons with an uncertainty band
4. **Classify** the chance of crossing the threshold within 60 minutes
5. **Explain** which features moved this particular prediction
6. **Persist** the prediction with timestamp, model version and data quality
7. **Push** the result to the console over a WebSocket

Training and serving call the same feature engineering module, so what the model was fitted on is what it sees in production.

Data

Synthetic by design, and said so plainly

Candidate research datasets need signed data use agreements, so they cannot ship in a public repository or be fetched during a build. GlycoTwin generates its cohort instead, which makes the whole pipeline reproducible from a clean checkout.

Patients	Observations	Mean glucose	Above 180
40	161,280	155	19.8%
14 days each	5 minute sampling	mg/dL	of all readings

The cost is real and we state it: every metric in this deck measures how well the model recovers a simulator written for this project, not human physiology. The brief warns that Type 1 research data cannot establish validity for Indian Type 2 patients. A synthetic cohort establishes even less.

Clinical target

The target was fixed before training

Threshold	Event horizon	Forecast	Event rate
180	60	30 / 60 / 120	11.8%
mg/dL, top of time in range	minutes	minutes ahead	in the scored population

The decision that most affects the numbers: the event is scored **only on readings at or below the threshold**. Scoring every reading would include ones already above it, where the answer is close to trivial and the metrics flatter the model.

Low glucose is not predicted. The cohort holds too few readings below 70 mg/dL to support a hypoglycaemia model, so none was trained.

competition.txt sections 4 and 13

How it was evaluated

Two protocols, and a baseline that has to be beaten

Patient level holdout, primary

40 patients split 26 / 6 / 8. Test patients appear in no part of training, early stopping, calibration or threshold selection.

Chronological holdout, secondary

Retrained without the final 3 days, then scored on those days, so no future reading leaks backwards.

Baselines on identical rows

- **Persistence:** future glucose equals the latest reading
- **Ridge regression** on the same features
- **Slope extrapolation** for the event

Glucose is strongly autocorrelated. A model that cannot beat persistence is not doing anything, which is why it is reported beside every figure.

Results

Forecast accuracy on patients the model never saw

Horizon	MAE	RMSE	Persistence MAE	Ridge MAE	Gain over persistence
30min	6.269	8.437	10.645	8.044	41.11%
60min	11.439	16.471	17.265	13.09	33.74%
120min	16.119	22.229	24.883	17.156	35.22%

The chronological holdout agrees: 5.413, 10.142 and 14.422 mg/dL at the three horizons. Two different ways of splitting the data give the same picture, so the gain is not an artefact of one split.

Per patient MAE at 60 minutes ranges from 10.02 to 14.121 mg/dL. Pooled averages hide that spread, so both are reported.

All figures from evaluation/metrics.json, written by the training run

Results

Catching a crossing before it happens

Metric	Model	Slope baseline
Precision	0.7951	0.4891
Recall (sensitivity)	0.5592	0.3476
Specificity	0.9806	0.9512
PR-AUC	0.7619	0.4237
ROC-AUC	0.9448	0.7729
Brier score	0.0527	0.0972

Scored on 27,168 readings at or below the threshold, operating point 0.34 chosen on validation patients. Expected calibration error 0.0179, so a stated 70% means close to 70% in practice.

Recall is 56% at this operating point, so roughly four crossings in nine are missed. The threshold is adjustable, and the next slide shows what moving it costs.

Confusion matrix: 1800 TP, 464 FP, 1419 FN, 23485 TN

The operating point

Where the alert line goes is a clinical choice, not a metric

The training run picks a default by F1 on validation patients. That is a statistical default. It says nothing about whether a missed crossing or a false alarm is worse, so the console hands the choice over and shows what it costs.

Threshold	Crossings caught	Alert episodes per patient per day	True alerts	Median warning
0.1	88%	7.8	41%	25 min
0.2	68%	4.8	60%	25 min
0.35 (default)	56%	3.5	73%	25 min
0.5	50%	3.1	80%	30 min
0.8	36%	2.2	93%	25 min

Burden is counted as **episodes**, not flagged readings. Readings arrive every 5 minutes, so one rising episode trips the threshold repeatedly; counting each firing separately overstates what reaches a clinician by roughly six times. Firings within 60 minutes are treated as one alert.

Median warning time hardly moves across the range, so a lower threshold buys more caught crossings rather than earlier ones. No setting is free, and nobody has yet asked a clinician where the line belongs.

Measured over 112 held out patient days

An honest result

The uncertainty band does not fully deliver

Intervals come from quantile regression at 0.10 and 0.90. On unseen patients the raw quantiles covered well under the nominal 80%, so conformal calibration fitted on the validation patients widens them.

Horizon	Raw quantile coverage	After conformal	Mean width
30min	0.690	0.765	19.1 mg/dL
60min	0.654	0.758	30.4 mg/dL
120min	0.652	0.749	43.2 mg/dL

Calibration helps but does not close the gap, because a previously unseen patient breaks the exchangeability the conformal guarantee assumes. **Bands shown in the console are narrower than they should be.** We report this rather than quoting the nominal 80%.

Romano, Patterson and Candes, Conformalized Quantile Regression, 2019

The console

What the clinician sees

- Measured readings and model output drawn as two separate series, never merged
- Forecast curve with its uncertainty band and the threshold line
- Risk panel in plain language, with projected crossing time
- Per prediction feature contributions, raising and lowering
- Data quality: gaps, stale readings, rejected values, late arrivals
- Replay, simulate, pause, restart, with a speed control
- Prediction history, every forecast the twin has produced
- Twin timeline pairing past forecasts with what was later measured
- Hypothetical scenarios, labelled hypothetical everywhere
- A results page reading metrics.json as produced

A persistent banner states what the system is not. Rejected readings are counted and surfaced, never quietly replaced with an invented value.

What-if, carefully

Scenarios show a model response, not a treatment effect

Four scenarios re-run the model on a modified recent window: a brisk walk, a lighter last meal, a later last meal, more sleep. Two guards sit on the feature.

1. **Magnitudes stay inside the range the cohort contains.** Gradient boosted trees do not extrapolate, so an input pushed past the largest split the model learned produces a number that stops responding and means nothing.
2. **Every modified input is checked against the training distribution** and reported as within range, extreme, or outside it. The console shows that check rather than hiding it.

Effects come out small, usually a few mg/dL, because recent glucose history dominates. Activity and time of day also move together in this cohort, so the response to an activity change reflects that association. We show the measured result and say what it is not.

Limitations

What this prototype does not establish

1. The data is synthetic. Performance here predicts nothing about real CGM data.
2. No claim whatever about Indian Type 2 patients. No prospective validation, no clinical review, no real population.
3. Prediction intervals under cover on unseen patients.
4. Recall is moderate at the default. It is adjustable, but no setting is free: catching 88% of crossings costs about 8 alert episodes per patient per day, most of them false.
5. Synthetic EHR demonstrates integration, not a realistic case mix.
6. Scenario outputs are associations learned from a generator, not causal effects.
7. No hypoglycaemia prediction.
8. Time to threshold is interpolated off the forecast curve, not a time to event model.

Future work

Where it goes next

Data

- Repeat the evaluation on a licensed CGM dataset under a data use agreement
- Prioritise a Type 2 cohort, and report the domain gap where one is used
- Per patient calibration using only earlier observations

Model and system

- Close the interval coverage gap with per patient conformal calibration
- Set the operating point with clinical input, using the measured burden
- Drift detection as a patient's profile changes
- An automated test suite, which this prototype does not have

> A credible twin is one where the data, the predictions, the evolving state and the evaluation hold together, and where the gaps are named.